1. 3rvrbefb

2. Evidence Acquisition and Integrity

Field	Value
Original file name	History
Source	Assessment Case Materials
File type	Chrome History file
SHHA-256 (original)	b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19
SHA -256 (working copy)	
Analysis System	
Note	

2.1 Case folder Structure

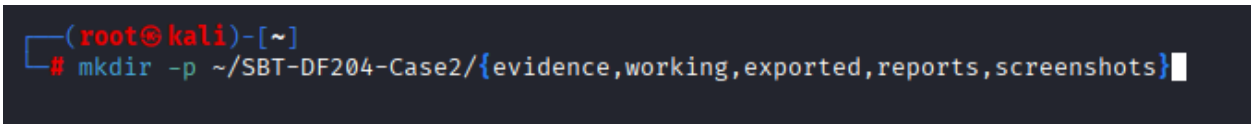


Image 1 (EO1). mkdir -p ~/SBT-DF204-Case2/{evidence,working,exported, reports and screenshots} – creates the subfolders for evidence and file storage throughout the studies.

2.2 Downloading evidence file

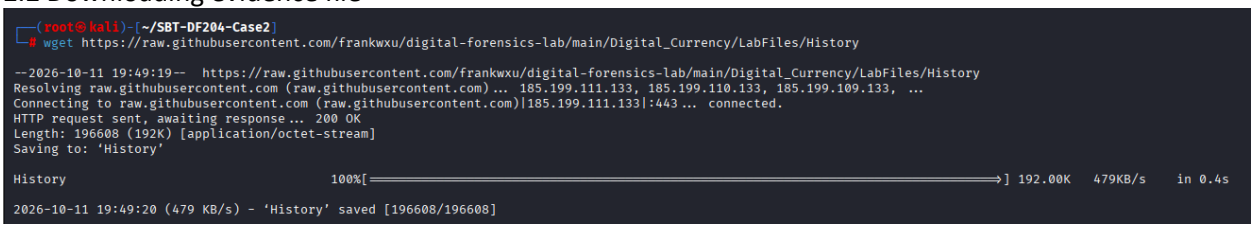


Image 2 (EO2). The file History was successfully downloaded and saved

2.3 Verifying Integrity

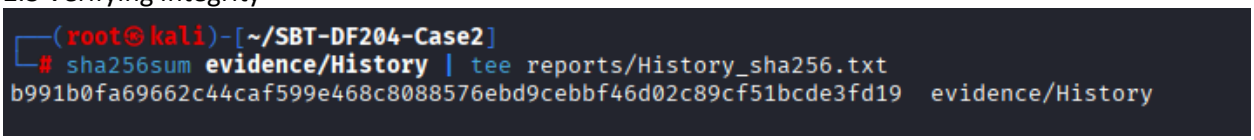


Image 3 (EO3). SHA256sum evidence/History | tee reports/History_Sha256.txt, recorded the SHA256 hash of the original evidence file

2.4 Creating a working Path

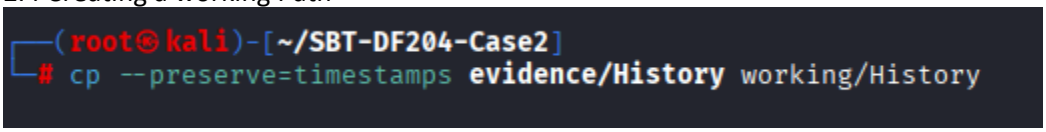


Image 4 (E04). Cp –preserve=timestamps evidence/History working/History. A working part is created to preserve integrity

3. Examining the file (History)

```
┌──(root㉿kali)-[~/SBT-DF204-Case2]
└─# file working/History
working/History: SQLite 3.x database, last written using SQLite version 3036000, file counter 33, database pages 41, cookie 0×a7, schema 4, UTF-8, version-valid-for 3
3
```

Image 5 (E05). File working/History, checking the file type and tables

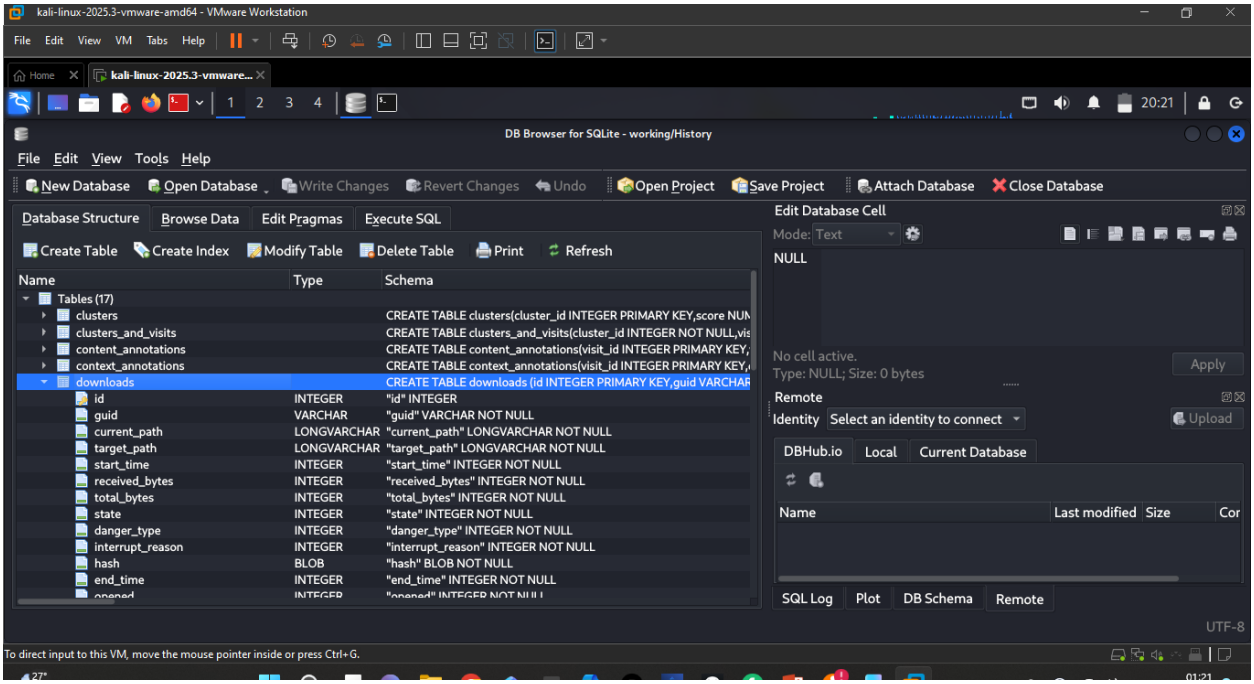


Image 6 (E06). Showing the Downloads Structure which is to save location and other download details.

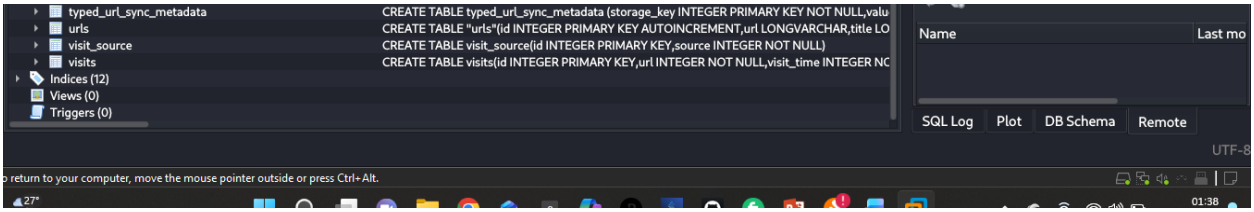


Image 7 (E07). Showing the urls which Stores visited URLs, including page titles and when was the last time the URL was visited. (and) Visits structure that Stores each visit to a URL, including when the URL is visited and how long a user stay on the page.

4. Building the Core Timeline

Running the SQL Query (SELECT
v.id AS visit_id,
u.id AS url_id,
u.url,
u.title,
datetime(
v.visit_time/1000000 +
strftime('%s','1601-01-01'),

'unixepoch'
) AS visit_utc,
v.from_visit,
v.transition,
v.visit_duration/1000.0 AS duration_ms
FROM visits v
JOIN urls u
ON v.url=u.id
ORDER BY v.visit_time;) to have a primary timeline

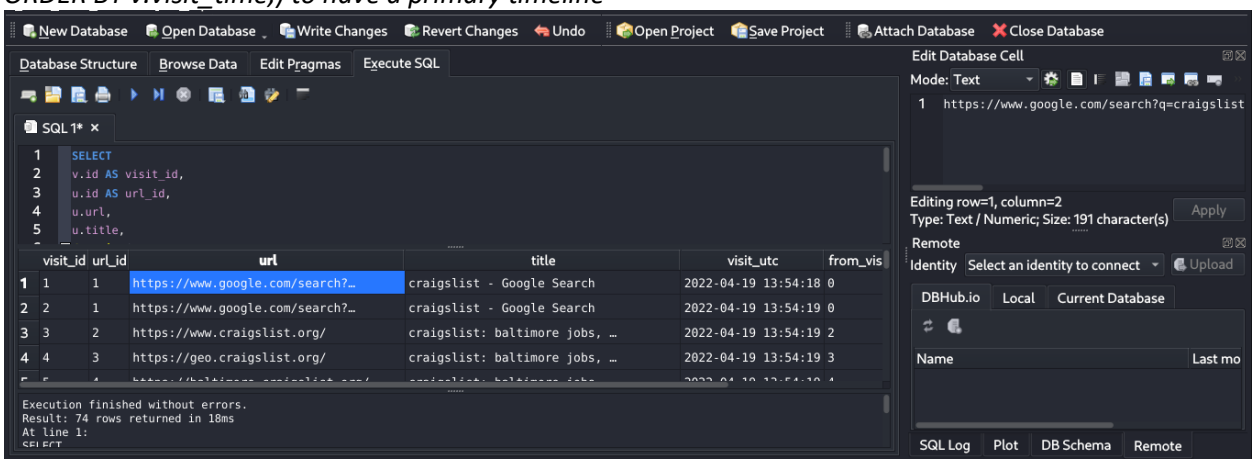


Image 6 (E06) indicates the above query implemented with results of 74 rows returned in 18ms at line 1

4.1 Analysis Strategy

Segment visit records based on visit duration (Labeling each segment as 1,2,…..)

Seg 1: visit Id = 1-2 (“search Craiglist”)

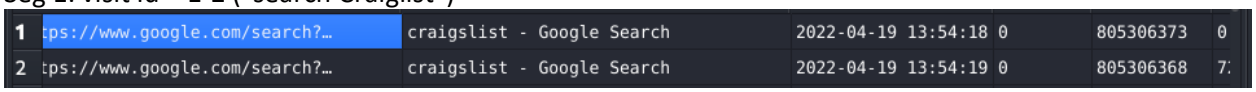


Image 7 (E07)

Visit ID	Date & Time	Description	Duration (in seconds)
1	2022-04-19 13:54:18	User typed “Craiglist” into chrome address bar and generate a url	O seconds which indicate a redirect
2	2022-04-19 13:54:19	Without any delay (0 second duration), chrome sends the URL to Google and google returns results in web page with the title “Craiglist” – Google search	User stayed in the page for 7 seconds

Seg 2 : visited ID = 3-5 (visited https://www.craigslist.org)

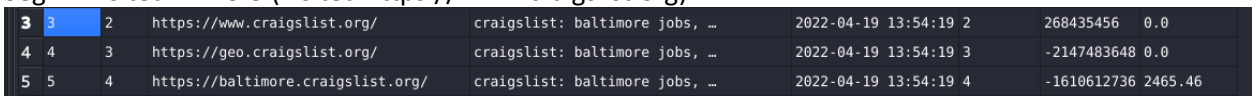
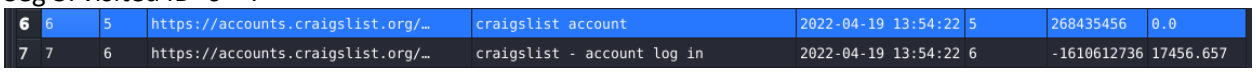


Image 8 (E08)

Visit ID	Date & Time	Description	Duration (in seconds)

3 - 5	2022-04-19 13:54:19	Suspect clicked the link (from visits id 2) and visited https://www.craiglist.org/. however, it was redirected to https://geo.craigslist.org/ and https://baltimore.craigslist.org/	Suspect stayed in the page https://baltimore.craigslist.org/ for 2.46546 seconds

Seg 3: visited ID=6 – 7



6	6	5	https://accounts.craigslist.org/…	craigslist account	2022-04-19 13:54:22	5	268435456	0.0
7	7	6	https://accounts.craigslist.org/…	craigslist - account log in	2022-04-19 13:54:22	6	-1610612736	17456.657

Image 9 (E09)

Visit ID	Date & Time	Description	Duration (in seconds)
6 - 7	2022-04-19 13:54:22	The Suspect click and account link https://accounts.craiglist.org/login/home to login and was redirected to another link https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L	Suspect stayed in the page https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L for 17.456 seconds

Seg 4: visited ID = 8

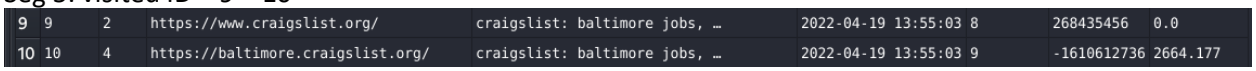


8	8	5	https://accounts.craigslist.org/…	craigslist account	2022-04-19 13:54:39	7	805306375	23439.248

Image 10 (E10)

Visit ID	Date & Time	Description	Duration (in seconds)
8	2022-04-19 13:54:39	The Suspect click and account link https://accounts.craigslist.org/login/home where the user filled out the login form and submit	Suspect stayed in the page 23 seconds

Seg 5: visited ID = 9 – 10

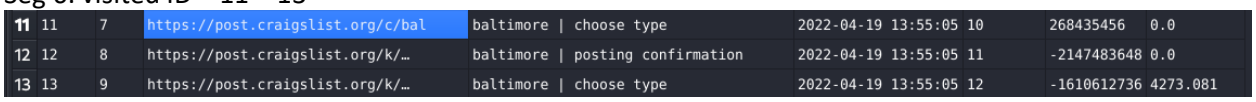


9	9	2	https://www.craigslist.org/	craigslist: baltimore jobs, …	2022-04-19 13:55:03	8	268435456	0.0
10	10	4	https://baltimore.craigslist.org/	craigslist: baltimore jobs, …	2022-04-19 13:55:03	9	-1610612736	2664.177

Image 11 (E11)

Visit ID	Date & Time	Description	Duration (in seconds)
9 - 10	2022-04-19 13:55:03	The user clicked the link https://www.craigslist.org/ and was redirected to https://baltimore.craigslist.org/	The user stayed on the page for 2.6 seconds

Seg 6: visited ID = 11 – 13



11	11	7	https://post.craigslist.org/c/bal	baltimore | choose type	2022-04-19 13:55:05	10	268435456	0.0
12	12	8	https://post.craigslist.org/k/…	baltimore | posting confirmation	2022-04-19 13:55:05	11	-2147483648	0.0
13	13	9	https://post.craigslist.org/k/…	baltimore | choose type	2022-04-19 13:55:05	12	-1610612736	4273.081

Image 12 (E12)

Visit ID	Date & Time	Description	Duration (in seconds)
11 - 13	2022-04-19 13:55:05	The user clicked the link https://post.craigslist.org/c/bal and was redirected to https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5 User was also redirected to https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=type	The user stayed on the page for four (4) seconds

Seg 7: visiting ID = (14, 15, ...20)

	visit_id	url_id	url	title	visit_utc	from_visit	transition	duration_ms
14	14	10	https://post.craigslist.org/k/…	baltimore | choose category	2022-04-19 13:55:10	13	805306375	5599.315
15	15	11	https://post.craigslist.org/k/…	baltimore | create posting	2022-04-19 13:55:15	14	805306375	177674.775
16	16	12	https://post.craigslist.org/k/…	baltimore | add map	2022-04-19 13:58:13	15	805306375	4389.706
17	17	13	https://post.craigslist.org/k/…	baltimore | create posting	2022-04-19 13:58:17	16	805306375	1649.042
18	18	14	https://post.craigslist.org/k/…	baltimore | create posting	2022-04-19 13:58:19	17	805306375	3199.944
19	19	8	https://post.craigslist.org/k/…	baltimore | posting confirmation	2022-04-19 13:58:22	18	805306375	4989.41
20	20	15	https://post.craigslist.org/manage/…	baltimore | manage posting	2022-04-19 13:58:27	19	805306368	5597.428

UTF-8

Image 13 (E13). Showing six (6) visited Id's with the urls, titles, utc, transitions and durations

Visit ID	Date & Time	Description	Duration (in seconds)
14	2022-04-19 13:55:10	The user clicked the link https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=cat	The user stayed on the page for five (5) seconds
15	2022-04-19 13:55:15	The user clicked the link https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=edit	The user stayed on the page for 177 seconds
16	2022-04-19 13:58:13	The user clicked the link https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=geoverify	The user stayed on the page for 4 seconds
17	2022-04-19 13:58:17	The user clicked the link https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=editimage	The user stayed on this page for 1.4 seconds
18	2022-04-19 13:58:19	The user under this ID clicked the link https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=preview	The user stays on the page for 3.1 seconds
19	2022-04-19 13:58:22	The user clicked this link https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5	And stays on this page for approximately 5 seconds
20	2022-04-19 13:58:27	The user follows this link https://post.craigslist.org/manage/7473121658	The user stays on the page for 5 seconds

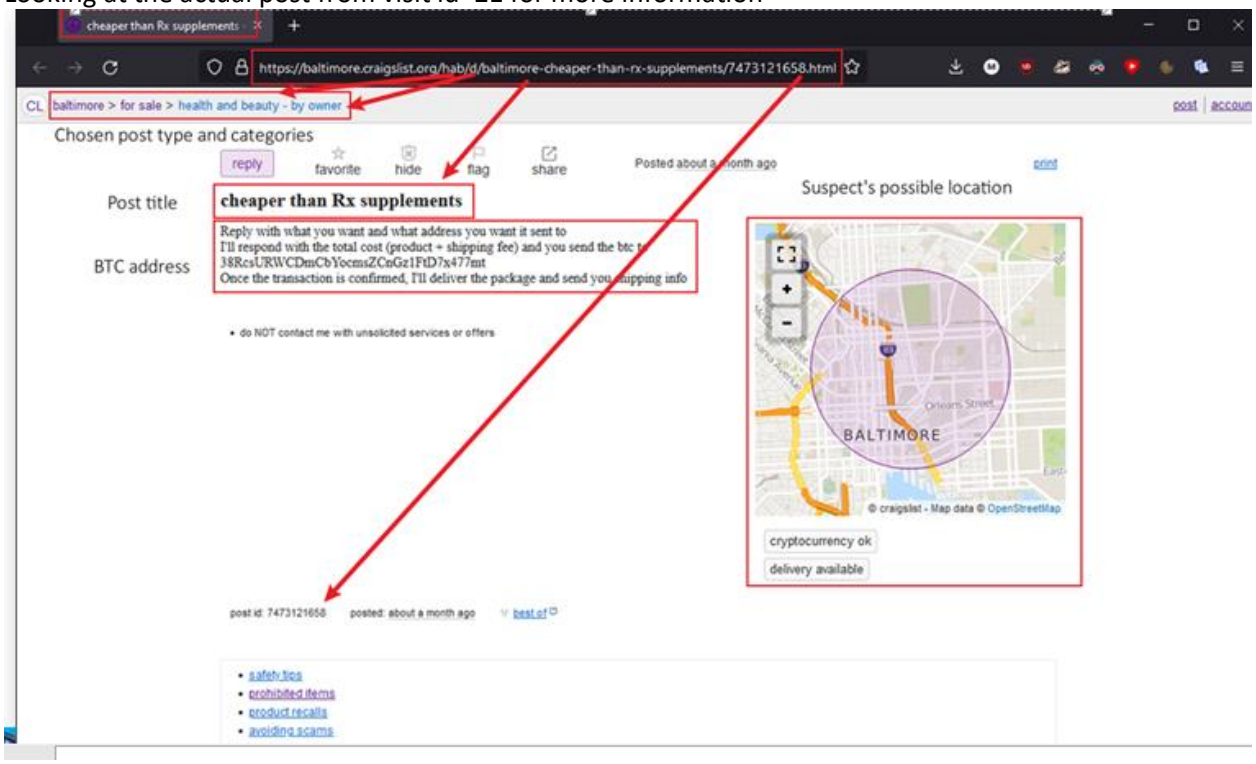
Seg 8: Visiting ID's = 21 – 23

	visit_id	url_id	url	title	visit_utc	from_visit	transition	duration_ms
21	21	16	https://baltimore.craigslist.org/hab…	cheaper than Rx supplements - health…	2022-04-19 13:58:31	0	805306368	0.0
22	22	17	https://www.google.com/search?…	gmail - Google Search	2022-04-19 13:58:38	0	838860805	0.0
23	23	17	https://www.google.com/search?…	gmail - Google Search	2022-04-19 13:58:39	0	805306368	842.727

Image 14 (E14)

Visit ID	Date & Time	Description	Duration (in seconds)
21 - 23	2022-04-19 13:58:31	The user clicked the link https://baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-supplements/7473121658.html) with path /hab/d/Baltimore-cheaper-than-rx-supplement to view post 747341212658 Then the user typed “Gmail” in chrome address bar and URL was generated to Google Search results	The user stayed on the page for 8. seconds

Looking at the actual post from visit id=21 for more information



Seg 9: visiting ID’s = 24 – 26

24	24	18	https://www.google.com/gmail/	Gmail: Free, Private & Secure Email …	2022-04-19 13:58:40	23	268435456	0.0
25	25	19	https://mail.google.com/mail/	Inbox (12) - unsub.fscs@gmail.com - …	2022-04-19 13:58:40	24	-2147483648	0.0
26	26	20	https://mail.google.com/mail/u/0/	Inbox (12) - unsub.fscs@gmail.com - …	2022-04-19 13:58:40	25	-2147483648	0.0

Image 15 (E15)

Visited ID	Date & Time	Description	Duration (in seconds)
24 - 26	2022-04-19 13:58:40	The user clicked the link https://www.google.com/gmail/ to Gmail then was	The user stayed on the page for 0 seconds

		redirectd to Inbox (12) - unsub.fscs@gmail.com - Gmail	

Seg 9: visited ID's = 27 – 29

27	27	21	https://accounts.google.com/…	Gmail: Free, Private & Secure Email …	2022-04-19 13:58:40	26	-2147483648	0.0
28	28	22	https://mail.google.com/intl/en/mail…	Gmail: Free, Private & Secure Email …	2022-04-19 13:58:40	27	-2147483648	0.0
29	29	23	https://www.google.com/intl/en/mail/…	Gmail: Free, Private & Secure Email …	2022-04-19 13:58:40	28	-2147483648	0.0

Visited ID	Date & Time	Description	Duration (in seconds)
27 - 29	2022-04-19 13:58:40	The user was redirected to accounts.google.com with the URL to their inbox as a parameter to ServiceLogin?service with https://mail.google.com/mall/u/O, User was then directed on ro an about.html page User was then redirected again to another sbout page with path /help/ to an about.html page	The user stayed on the page for 0 seconds

Seg 10: Visited ID's = (30, 31....36)

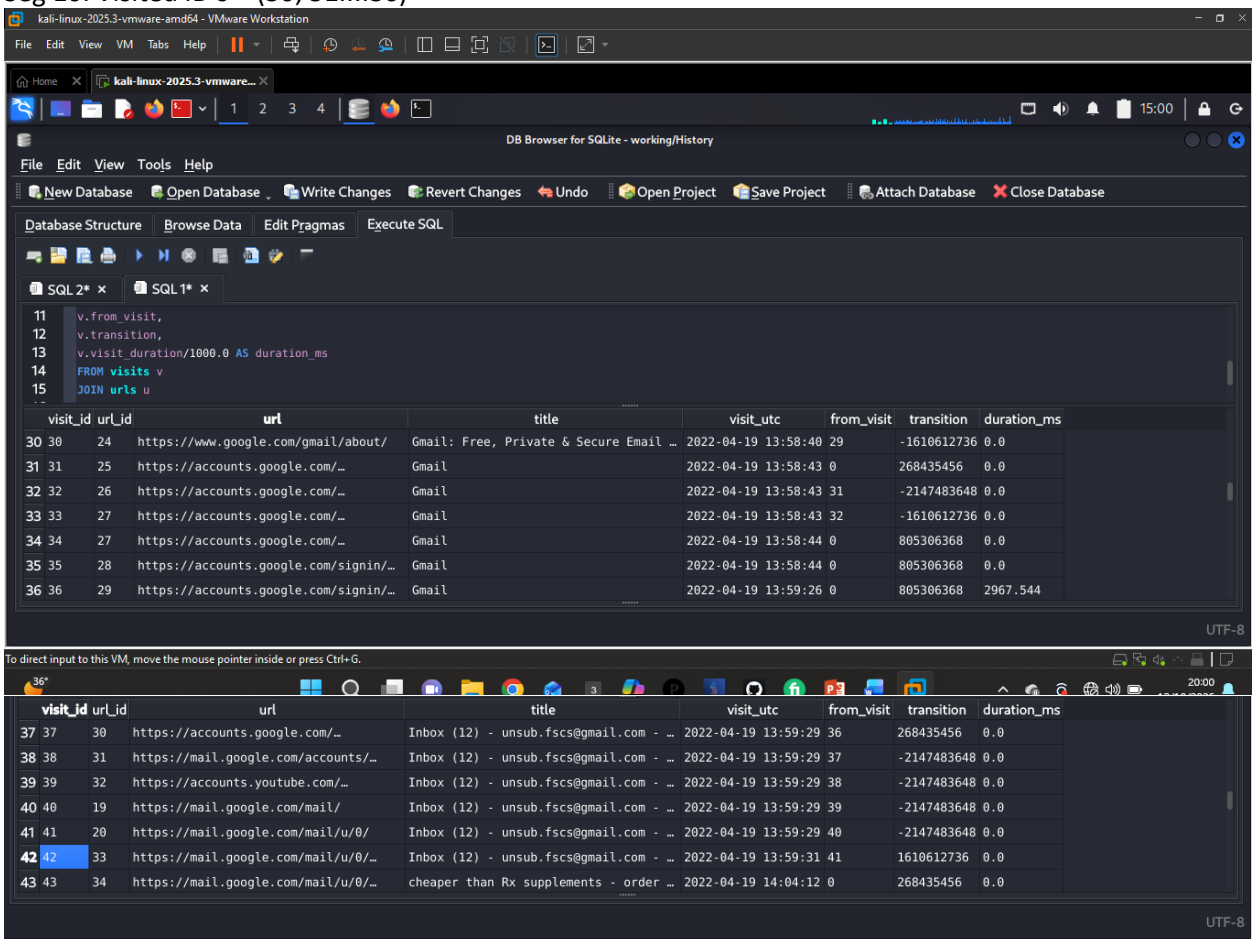
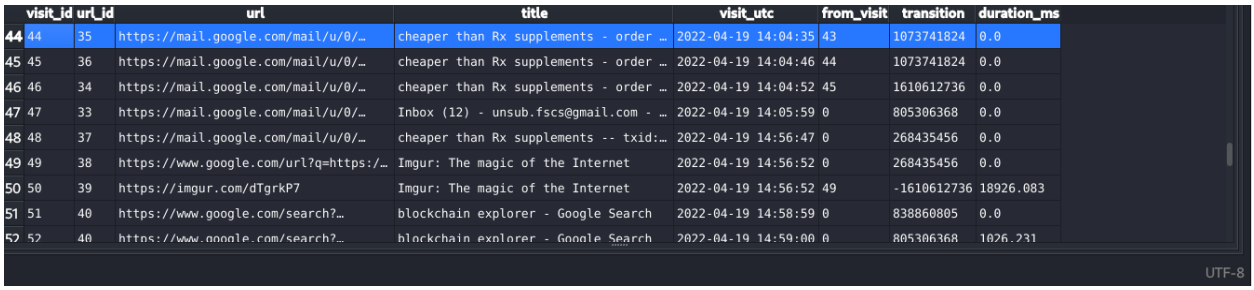


Image 16 (E16)

Visited ID	Date & Time	Description	Duration (in seconds)
30 - 32	2022-04-19 13:58:40	User clicked a link https://www.google.com/) withb path /gmail/about/ to Google Workspace then user click a link https://accounts.google.com/| with path /accounts Chooser/ and query siginChooser?service=mail to a login page Then user was redirected to link (https://accounts.google.com) with query AccountChooser?service=email to choose an account	The user stayed on the page for 0 seconds
33 - 36	2022-04-19 13:58:43	The user clicks on the web on the url (https://account.google.com) gmail with query ServiceLogin?continue= to a login page Then User clicked a link https://accounts.google.com/o to Gmail with same query (https://accounts.google.com) to Gmail with path /signin/ with query identifier?continue to a login page prompting identifier email address/username Then user clicked a link (https://account.google.com/) with path /sigin/challenge/ and query pwd?continue= to a login page prompting challenge password	Time spend is on this pages is approximately 3 sconds
37 - 39	2022-04-19 13:59:29	User clicked the link (https://account.google.com) with query CheckCookie?continue= and parameter value loginDonetHtml to unsub.fscs@gmail.com's inbox Then the user was redirected to (https://mail.google.com/) with path /accounts/ and query SetOSID?authuser to unsub.fscs@gmail.com's inbox and then the user was redirected (https://accounts.youtube.com/) with path /accounts/ and query SetSID? To unsub.fscs@gmail.com's inbox	The user stays on the page for 0 seconds
40 - 41	2022-04-19 13:59:29	User clicked a link (https://mail.google.com/) with path mail to unsub.fscs@gmail.com's inbox User then click a link (https://mail.google.com) with path /u/o/ to unsub.fscs@gmail.com's inbox	The user stays on the page for 0 seconds
42 - 43	2022-04-19 13:59:31	User clicked on the url https://mail.google.com/mail/u/0/#inbox to unsub.fscs@gmail.com's inbox User clicked a link https://mail.google.com/ with string WhctKKXXDrfCqXBGCnHkHGxZnZbkXKdXnJTmxwZwPtj VmvHMrfPPRthPXJgXngHcFLmggWv. Indicating ID for opened email with subject header cheaper than Rx supplemnts - order	User stayed for 0 seconds on that page

Seg 11: visiting ID's = (43, 44.....52)



	visit_id	url_id	url	title	visit_utc	from_visit	transition	duration_ms
44	44	35	https://mail.google.com/mail/u/0/…	cheaper than Rx supplements - order …	2022-04-19 14:04:35	43	1073741824	0.0
45	45	36	https://mail.google.com/mail/u/0/…	cheaper than Rx supplements - order …	2022-04-19 14:04:46	44	1073741824	0.0
46	46	34	https://mail.google.com/mail/u/0/…	cheaper than Rx supplements - order …	2022-04-19 14:04:52	45	1610612736	0.0
47	47	33	https://mail.google.com/mail/u/0/…	Inbox (12) - unsub.fscs@gmail.com - …	2022-04-19 14:05:59	0	805306368	0.0
48	48	37	https://mail.google.com/mail/u/0/…	cheaper than Rx supplements -- txid:…	2022-04-19 14:56:47	0	268435456	0.0
49	49	38	https://www.google.com/url?q=https:/…	Imgur: The magic of the Internet	2022-04-19 14:56:52	0	268435456	0.0
50	50	39	https://imgur.com/dTgrkP7	Imgur: The magic of the Internet	2022-04-19 14:56:52	49	-1610612736	18926.083
51	51	40	https://www.google.com/search?…	blockchain explorer - Google Search	2022-04-19 14:58:59	0	838860805	0.0
52	52	40	https://www.google.com/search?…	blockchain explorer - Google Search	2022-04-19 14:59:00	0	805306368	1026.231

UTF-8

Image 17 (E17)

Visited ID	Date & Time	Description	Duration (in seconds)
44 - 45	2022-04-19 14:04:35	The user clicked the link (https://www.google.com/gmail/) with the query ?compose=JHrtff referring to open message WhtcKKXX with subject header “order” User then clicked a link (https://mail.google.com) with query ?compose=DXDWs referring to same message WhcKKXX indicating two replies were made for the same message	The user stayed on the page for 0 seconds
46 - 48	2022-04-19 14:04:52	User clicked on the link https://mail.google.com/mail/u/0/#inbox/ with the strings WhctKKXXDrfCqXBGC “order” message open User click on the link https://mail.google.com/mail/u/0/#inbox/ with strings WhctKKXXDrfCqvtRj, another new message with subject header “txid: 517b2159…..”	User datyed in the page for 0 seconds
49 - 50	2022-04-19 14:56:52	User clicked a link (https://www.google.com/) with query url?=https://imgur.com/dtgrkp7 and parameter source=gmail indicating the imgur link originated from Gmail message User clicked a link (https://imgur.com/dtgrkp7) to imgur post	User stayed on the page for 18 seconds
50		Looking at the imgur post – URL does not have enough information about the image post Message shows Bitcoin was sent Amount of BTC sent and USD equivalent at time Suspect’s wallet address from craigslist post (38RcsURWCDmCBYocmsZCnGz1FtD7x477mt) And another link for block chain explorer (E18) below shows more details.	

Imgur is an online image sharing and hosting service. Images can be uploaded anonymously and shared. In database, you can see the imgur link from message has parameter source=gmail

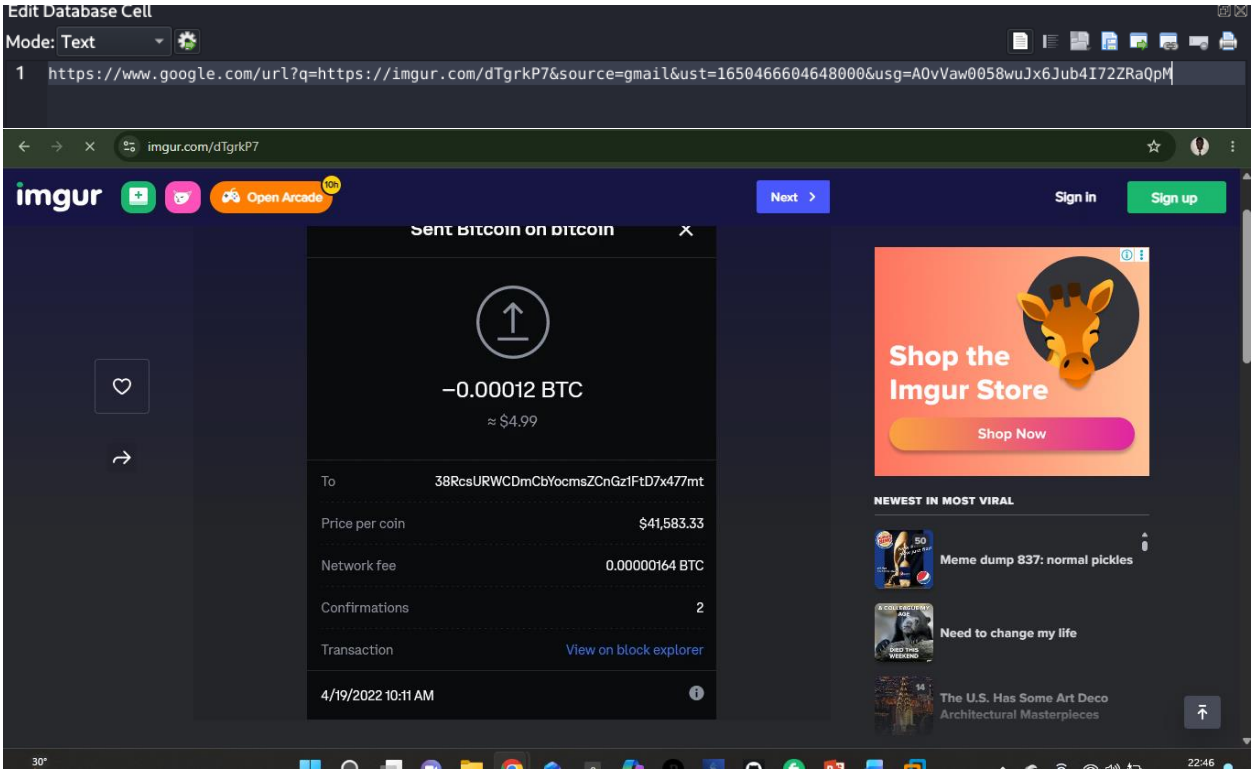


Image 18 (E18).
Blockchain Explorer is a block and crypto transaction search engine – queries can include wallet addresses and transaction IDs